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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded

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1 Kernel

1.1 Introduction

A piece of code that helps us to execute multiple programs to operating system by incorporating with hardware is kernel. It is a core part of an operating system which serves as the main interface between the computer's physical hardware and the processes running on it. The kernel enables multiple applications to share hardware resources by providing access to CPU, memory, disk I/O, and networking. (DigitalOcean, 2021)

1.2 Objective

- To establish communication between user-level applications and hardware.
- To learn in depth about kernel.
- To know the use of kernel in operating system
- To know the purpose of kernel.
- To control task management. (GeeksforGeeks, 2024)

1.3 Types of Kernel

There are five types of Kernel:

1.3.1 Monolithic:

It is an operating system where all the services of operating system run in kernel space which means they all share the same memory space. As they run in a same space, there is no overhead associated with moving data between kernel space and user space, which makes it generally faster. Examples of monolithic kernelbased operating systems include Unix, Linux, Open VMS, XTS-400, and z/TPF.

(GeeksforGeeks, 2024)

1.3.2 Microkernel

It handles all system resources where the user and kernel services are implemented in distinct address spaces. User services are kept in user address space, while kernel services are kept in kernel address space. If any of the user services fails, the kernel service is unaffected along with the OS service because kernel and user services are isolated in micro kernel. (javatpoint, 2024).

1.3.3 Nanokernel

A nano kernel is an operating system architecture that provides the bare minimum services required for the system to work. It does only the most necessary tasks, like managing basic computer operations and allowing parts of the system to talk to each other. More complex tasks are delegated to other components of the operating system. (NordVPN, 2024).

1.3.4 Hybrid kernel

Hybrid kernel is the combination of both microkernel and monolithic kernel. It has design and speed of monolithic kernel whereas stability and modularity of microkernel.

Example : Windows NT, Netware, BeOS etc. (GeeksforGeeks, 2024)

1.3.5 Exokernel

It is the type of kernel which follows end-to-end principle. It has fewest hardware abstractions as possible. It allocates physical resources to applications.

Example : Nemesis, ExOS etc. (GeeksforGeeks, 2024)

1.4 Conclusion

This logbook helped me analyse and research on Kernel which expanded my knowledge in this given topic. I learned various things varying from websites to websites which developed my core understanding. We were tasked to introduce kernel, write its objectives, and explain its types. This gave me a synchronous order of understanding the given topic accordingly and slowly and gradually discover newer

upcoming things related to it. It helped me not just limit my knowledge in it but expand via research.

1.5 References

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